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DAISI Challenge: Singapore Edition

Round 1 opens: Wednesday, 16 September 2026

Round 1 submissions close: Tuesday, 6 October 2026

Demo Day: Tuesday, 27 October 2026

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Overview

The Databricks AI Social Impact (DAISI) Challenge invites student teams from Singapore's Institutes of Higher Learning to build data-driven solutions for the country's most pressing social challenges, using Databricks and open government datasets from data.gov.sg.

Teams pick one problem statement and build an end-to-end solution, from ingestion and processing through machine learning, analytics and a live dashboard demo. Ten teams are shortlisted for a two-week mentored sprint, three are chosen at Demo Day, and the winning team presents at the new Databricks Singapore office launch.

TEAMS

1–4 students from Singapore IHLs

COST

Free on Databricks Free Edition

ROUND 1

1-page concept or 3-slide pitch

Key dates

16 Sep	Launch at Data+AI World Tour Singapore, registration opens
24 Sep	Hands-on training: Get Started with Databricks: End to End, 7–10pm SGT (completed; not recorded — self-paced course available)
6 Oct	Round 1 submissions close, 11:59pm SGT (1-page concept note or 3-slide pitch, PDF)
7–9 Oct	Shortlisting
9 Oct	Top 10 teams announced, with details to Demo Day shared
12 Oct	Virtual build kickoff, mentors assigned
12–26 Oct	Two-week mentored build sprint
27 Oct	Demo Day, 4-7pm SGT, top 3 selected
4 Nov	Finale at the new Databricks Singapore office

Problem statements

Three tracks, nine statements. All datasets are freely available on data.gov.sg, most with no login required.

TRACK A

Healthy Nation: Caring for an Ageing Singapore

Singapore becomes officially "super-aged" in 2026 — 1 in 5 residents will be 65 or older. Social services, healthcare infrastructure and communities are under pressure to adapt faster than ever.

A1

SilverWatch: Predicting social isolation risk across Singapore

THE PROBLEM

Social isolation among the elderly is a silent health crisis — a stronger predictor of premature death than obesity or smoking. Outreach by the Silver Generation Office and Active Ageing Centres is largely manual and coverage is uneven. As the elderly population surges past 1 million, data-driven prioritisation of social care resources is critical.

WHAT TO BUILD

A data platform that ingests demographic and social care infrastructure data, models social isolation risk at HDB town and planning area level, and produces a decision-support tool for social workers and volunteer coordinators. Combine data engineering, a composite risk-scoring model, and a live BI dashboard that non-technical community workers can use.

AVAILABLE DATASETS

Dataset	Source	What it contains
Key Indicators on the Elderly (Annual)	SingStat	National elderly health, living and social indicators
HDB Elderly & Future-Elderly Population	HDB	Block-level elderly population by age and sex
Senior Activity Centres & Active Ageing Centres	MSF	Locations and capacity of community centres
Households Assisted Through ComCare Schemes	MSF	Households receiving social assistance by area
Resident Population 65+ by Living Arrangements	SingStat	Proportion living alone, with family, and so on

WHAT A WINNING DEMO LOOKS LIKE

- A geospatial heatmap of planning areas ranked by a composite isolation risk score
- An ML model combining elderly density, living-alone rates and AAC coverage gaps into a risk index
- A prioritised list of the top 10 planning areas for targeted outreach
- A dashboard a social worker or volunteer coordinator could actually use

STRETCH GOALS

- Resource allocation recommender: where should new AAC spots go to maximise coverage?
- Temporal projection: how isolation risk grows by 2030 on current demographic trends
- Governance layer: responsible data access controls in Unity Catalog

A2

DengueRadar: Real-time outbreak forecasting

THE PROBLEM

Dengue is hyperendemic in Singapore, with outbreaks affecting tens of thousands of residents. Rising temperatures and erratic rainfall extend the Aedes breeding season. NEA cluster monitoring today is largely reactive — predictive risk intelligence could let communities act 2–4 weeks ahead.

WHAT TO BUILD

A near-real-time pipeline combining live dengue cluster data, weather station readings and geospatial data into a forward-looking dengue risk forecast by planning area. Show streaming or incremental ingestion from live APIs, weather feature engineering, and a predictive model on an interactive map.

AVAILABLE DATASETS

Dataset	Source	What it contains
Dengue Clusters (GeoJSON)	NEA	Active cluster polygons with case counts
Dengue Cases by Region	NEA	Historical counts by sub-region
Rainfall across Singapore (Real-time API)	NEA	5-minute station rainfall readings

Dataset	Source	What it contains
Air Temperature across Singapore (Real-time API)	NEA	5-minute temperature readings island-wide
Historical Rainfall Collections (2016–2024)	NEA	Multi-year rainfall history for training

WHAT A WINNING DEMO LOOKS LIKE

- A real-time dashboard of current cluster intensity mapped across Singapore
- A 2-week risk forecast model trained on weather signals and historical cases
- Low / Medium / High risk classification per planning area with explainable features
- Evidence the model beats a baseline such as a recent-history average

STRETCH GOALS

- Layer in hawker centre density or green space as environmental risk proxies
- Automated alerts when risk crosses a threshold
- Track experiments in MLflow and compare feature sets

A3

HealthPulse: Community healthcare demand forecasting

THE PROBLEM

Under Healthier SG, MOH is shifting from hospital-centric to community-based care. But where should polyclinics, nursing homes and community health posts expand? Much planning relies on top-down projections rather than granular community demand signals — getting it wrong means wasted investment or critical care gaps.

WHAT TO BUILD

A forecasting platform projecting community healthcare demand at planning area level over a 5-year horizon. Build a reproducible pipeline across MOH and population datasets, apply time-series or regression forecasting, and surface a geospatial planning dashboard.

AVAILABLE DATASETS

Dataset	Source	What it contains
Hospital Admissions (Public Sector, Monthly)	SingStat / MOH	Monthly inpatient admissions and day surgeries
Health Facilities & Beds in Inpatient Facilities	MOH	Nursing homes, hospitals and beds by type
HDB Elderly Population by Planning Area	HDB	Age and sex-segmented elderly population by block
Total Population by Broad Age Group & Sex (Annual)	SingStat	National population distribution trends
Healthcare Schemes and Subsidies	MOH	Active government care programmes by scheme

WHAT A WINNING DEMO LOOKS LIKE

- A 5-year demand forecast for community care at planning area level
- Supply vs. projected demand gap analysis in an interactive dashboard
- A map of the top 5 underserved areas needing infrastructure investment
- A clear methodology policymakers could trust and act on

STRETCH GOALS

- Policy impact modelling: demand reduction if Healthier SG hits prevention targets
- Unity Catalog access policy reflecting health data sensitivity
- Scenario toggles for optimistic, base-case and pessimistic ageing

TRACK B

Green & Resilient City: Climate Action for Singapore

2025 was Singapore's hottest year and wettest March on record, and 2026 is the Year of Climate Adaptation. The data is real, the urgency is now, and the open datasets are live.

B1

HeatGuard: Urban heat risk mapping for vulnerable communities

THE PROBLEM

Urban heat islands are intensifying across dense residential estates. Elderly residents in lower-floor HDB flats — many without air-conditioning or facing high utility bills — are at greatest risk. Heat stress leads to excess hospitalisations, lost productivity and preventable deaths among the most vulnerable.

WHAT TO BUILD

A real-time heat monitoring and risk intelligence platform combining island-wide temperature sensor data with demographics to identify the planning areas and blocks at highest risk and support proactive community alerts.

AVAILABLE DATASETS

Dataset	Source	What it contains
Air Temperature across Singapore (Real-time API)	NEA	5-minute readings across ~50 weather stations
PM2.5 Air Quality (Real-time)	NEA	15-minute particulate readings by region
HDB Elderly Population by Planning Area	HDB	Elderly density by block and planning area
Rainfall across Singapore (Real-time API)	NEA	Station rainfall for heat-humidity correlation
Senior Activity Centres (locations)	MSF	Community cooling refuge points

WHAT A WINNING DEMO LOOKS LIKE

- A live heat index dashboard across the weather station network
- An ML “Heat Risk Score” by planning area (temperature + elderly density + cooling access)
- Automated Low / Medium / High classification with tiered alert recommendations
- At least two real-time data streams integrated

STRETCH GOALS

- Cooling infrastructure simulation for the highest-risk zones
- Green space analysis using parks and cycling corridors as a cooling proxy
- Next-day heat risk forecast from forecast APIs and historical patterns

B2

FloodSense: Flash flood prediction & urban drainage intelligence

THE PROBLEM

Singapore recorded its wettest March on record in 2025, with flash floods causing property damage, transport disruption and community distress. PUB drainage is extensive, but extreme rainfall is growing more frequent. Predictive flood intelligence would help residents, town councils and agencies prepare.

WHAT TO BUILD

A predictive flash flood risk model using real-time and historical rainfall, flood-prone area records and live flood alerts to forecast which areas are most at risk and how that risk evolves. Build a training set from historical rainfall and flood co-occurrence, train a classifier, and wire live streams to score current risk.

AVAILABLE DATASETS

Dataset	Source	What it contains
Rainfall across Singapore (Real-time API)	NEA / PUB	5-minute rainfall readings from the station network
Historical Rainfall Collections (2016–2024)	NEA	Multi-year data for model training
Flood Alerts across Singapore (Real-time API)	PUB	Live flood alert events
Flood Prone Areas (Annual)	PUB	PUB-designated flood-prone hectares, 2022–2025

WHAT A WINNING DEMO LOOKS LIKE

- An ML classifier predicting Low / Medium / High flood risk by zone
- A live risk dashboard updating as real-time rainfall streams in
- A 3-year time-series trend analysis of flood-prone areas
- Both historical batch data (training) and real-time streaming data (inference)

STRETCH GOALS

- Cascading impact: correlate flood events with LTA ridership disruption
- Climate projection layer using sea-level rise scenarios
- Uncertainty quantification with confidence intervals

B3

ZeroWaste Navigator: Singapore's circular economy intelligence platform

THE PROBLEM

Singapore generates over 7 million tonnes of waste annually. The Zero Waste Masterplan targets a 30% reduction by 2030 and Semakau landfill has a finite runway. Household recycling rates stay stubbornly low — partly because infrastructure coverage is uneven and contamination is high.

WHAT TO BUILD

A platform analysing the waste and recycling landscape, identifying planning areas underserved by recycling infrastructure, and modelling the impact of targeted interventions on national recycling rates — geospatial analysis, trend modelling and scenario simulation in one dashboard.

AVAILABLE DATASETS

Dataset	Source	What it contains
Waste Disposed & Recycled (Annual)	SingStat / NEA	National waste tonnage and recycling totals
Waste Management & Overall Recycling Rates (Annual)	SingStat	National recycling rate time series
Recycling Bin Locations	NEA	Geospatial public bin locations
E-Waste Collection Points (GeoJSON)	NEA	E-waste drop-off point locations

Dataset	Source	What it contains
Hawker Centres (GeoJSON)	NEA	Food waste context — hawker centre locations
General Waste Collectors (Licensed)	NEA	Licensed collectors by waste class

WHAT A WINNING DEMO LOOKS LIKE

- Geospatial analysis of recycling bin coverage per 1,000 residents by planning area
- National waste and recycling trend analysis with an ML-projected trajectory to 2030
- The top 5 planning areas where investment would most improve recycling outcomes
- A clear visual "before vs. after intervention" comparison

STRETCH GOALS

- Route optimisation for collection logistics to cut carbon and cost
- Food waste estimation under hawker centre policy changes
- Recycling behaviour model to target campaigns by demographics

TRACK C

Liveable & Inclusive City: Housing, Mobility & Community

Affordable, connected and inclusive living is under pressure from rising HDB prices, transport accessibility gaps, and threats to the hawker culture that binds communities together.

C1

FlatFair: HDB resale market intelligence & affordability forecasting

THE PROBLEM

HDB resale prices have risen sharply over the past decade, creating real anxiety among young Singaporeans about homeownership. Despite one of the most data-rich housing

markets in the world, buyers navigate it with limited transparent analytics — and the 2024 flat classification change shifted market dynamics again.

WHAT TO BUILD

An end-to-end analytics platform on HDB resale transaction data surfacing price trends, affordability patterns and forward-looking forecasts by town and flat type — a full pipeline from raw transactions to an ML forecast, with a polished interactive dashboard.

AVAILABLE DATASETS

Dataset	Source	What it contains
Resale Flat Prices (Jan 2017 – present)	HDB	Town, flat type, floor area, storey, price — dataset ID d_8b84c4ee58e3cfc0ece0d773c8ca6abc
HDB Annual Report 2025	HDB	Supply pipeline, policy context, market overview
Total Population by Planning Area	SingStat	Demographic demand context

WHAT A WINNING DEMO LOOKS LIKE

- An interactive dashboard of price trends by town, flat type, storey range and year
- An ML forecast of resale prices for the next 6 months at town level
- An affordability index tracking prices against income benchmarks
- A production-quality pipeline handling nulls, outliers and schema evolution

STRETCH GOALS

- Fair value estimator: is a listed price above or below expected market value?
- BTO supply impact model on nearby resale transactions
- Unity Catalog governance with lineage and quality documentation

C2

MoveEasy: Mobility accessibility intelligence for inclusive communities

THE PROBLEM

Singapore ranks among the world's best public transport systems, yet accessibility gaps persist. Elderly residents, people with mobility challenges and underserved communities still struggle to reach hospitals, Active Ageing Centres and polyclinics. First and last-mile connectivity determines quality of life for the most vulnerable.

WHAT TO BUILD

A mobility analytics platform identifying transport accessibility gaps across Singapore, combining geospatial data with demographics and ridership to produce an evidence-based accessibility score by planning area.

AVAILABLE DATASETS

Dataset	Source	What it contains
Cycling Path Network (GeoJSON)	LTA	Current cycling path coverage
Master Plan 2025 Cycling Layers (GeoJSON)	URA	Planned future cycling network
Public Transport Operation & Ridership (Annual)	SingStat / LTA	Annual ridership by mode and route usage
Senior Activity Centres & Active Ageing Centres	MSF	Key community service destinations
Childcare Centre Listings (GeoJSON)	ECDA	Locations of childcare services
HDB Elderly Population by Planning Area	HDB	Elderly density for demand-weighted analysis

WHAT A WINNING DEMO LOOKS LIKE

- Distance from each area's elderly population centroid to the nearest AAC, polyclinic and cycling path
- An accessibility score by planning area in a ranked dashboard
- The 10 areas with the highest elderly density and lowest accessibility
- A map that communicates findings without technical expertise

STRETCH GOALS

- Infrastructure simulation for one new cycling path or bus route
- Master Plan 2025 overlay showing future accessibility gains
- Extend the lens to childcare accessibility for working parents

C3

KopilamAI: Hawker culture, food access & community sustainability

THE PROBLEM

Hawker centres are the beating heart of Singapore's social fabric — affordable, culturally diverse meals and de facto community gathering spaces. Rising costs, an ageing hawker workforce and shrinking supply threaten the institution, while hawker food waste adds to the national waste challenge.

WHAT TO BUILD

A platform analysing the distribution, accessibility and sustainability of the hawker network — identifying food access gaps and surfacing closure and food waste patterns that threaten hawker culture, told as an evidence-based story communities and policymakers could act on.

AVAILABLE DATASETS

Dataset	Source	What it contains
Hawker Centres (GeoJSON)	NEA	Locations, names, stall counts and attributes
Dates of Hawker Centre Closures	NEA	Historical renovation and permanent closure dates
Waste Disposed & Recycled (Annual)	SingStat / NEA	Food waste tonnage trends
Recycling Bins (Locations)	NEA	Infrastructure context for sustainability analysis
Total Population by Planning Area	SingStat	Demand-side denominator for coverage analysis
Resale Flat Prices by Town	HDB	Affordability pressure context by area

WHAT A WINNING DEMO LOOKS LIKE

A geospatial dashboard of hawker centre density per 10,000 residents by planning area

STRETCH GOALS

Site suitability model recommending locations for new hawker centres

Social vulnerability multiplier: high elderly concentration plus low hawker

A food access gap analysis for dense areas with low hawker coverage

A trend analysis of closures over time — permanent vs. temporary

A food waste estimation model tied to hawker activity and area demographics

density

Economic modelling of affordability impact if hawker density drops

Working with data.gov.sg

Most datasets can be downloaded directly as CSV or GeoJSON from the dataset page, queried through the datastore API, or streamed live for weather, air quality, flood alerts and bus arrivals.

```
https://data.gov.sg/api/action/datastore_search?resource_id=<dataset_id>&limit=100
```

Recommended Databricks workflow

1 **Data ingestion**

Ingest raw datasets from data.gov.sg APIs and CSVs into Delta Lake tables using Auto Loader or notebook-based ingestion with schema inference.

2 **Data engineering**

Clean, join and transform raw data into analysis-ready tables. Apply data quality checks and document lineage.

3 **Machine learning**

Build and train models on the prepared datasets. Use MLflow to track experiments, compare runs and register the best model.

4 **Analytics & BI**

Build an interactive AI/BI dashboard that makes findings accessible to a non-technical audience and clearly actionable.

5 **Governance (stretch)**

Catalogue datasets in Unity Catalog. Document sources, apply access controls appropriate to sensitivity, and show lineage from raw source to final model.

Databricks Free Edition

Free Edition is a no-cost, serverless workspace for learning and prototyping with data and AI. It includes usage quotas and is intended for non-commercial use.



01 **Create your account**

Sign up for Databricks Free Edition with your preferred sign-in method. An existing Free Edition account works too.

02 **Open your workspace**

Complete the welcome flow and confirm you can create a notebook, start the SQL warehouse and navigate the workspace.

03 **Set up the team**

Let every teammate explore in their own Free Edition workspace, then choose one workspace as the final demo workspace.

04 **Deploy early**

If you are building an app, get a small version running in the final workspace before adding features.

05 **Prepare the demo state**

Test the exact path you will show, keep required data in the final workspace, and record a fallback for any live step.

Create a Free Edition account

What you can build

An AI app or agent using Databricks Apps, Foundation Model APIs and model serving.

An analytics experience using notebooks, SQL, dashboards and Genie spaces.

A data pipeline that ingests, cleans and transforms data with Lakeflow.

A database-backed app using Lakebase and Databricks Apps.

An ML experiment tracked and evaluated with MLflow.

A practical build path

Define the user and problem in one sentence: who has it, what they need, how you help.

Prepare a small dataset you can load, query and reproduce before scaling up.

Build one end-to-end slice — data, core logic and user-facing output — before extras.

Add the Databricks capability that matters because it improves the solution, not to pad the feature count.

Deploy and rehearse from the final workspace, within the organiser's time limit.

Use AI and data responsibly

Never upload confidential, customer or personal data you do not have permission to use, or any credential.

Review AI-generated code, SQL, citations and outputs, and test the cases that matter to your demo.

Credit datasets, libraries, models and templates in your submission.

Be clear with judges when data is synthetic, a response is cached, or a feature is a prototype.

Limits & quotas to plan around

Free Edition is serverless-only and governed by fair-usage quotas. When a quota is exceeded, the affected compute can become unavailable for the rest of the day, and in extreme cases the rest of the month. Keep the project small and test the final flow before the demo.

SQL warehouse: one serverless warehouse at 2X-Small size.

Jobs: up to five concurrent job tasks per account.

Lakeflow pipelines: one active pipeline per supported pipeline type.

Model serving: limited active endpoints; no GPU serving and no provisioned throughput.

AI Search: one endpoint and one search unit; Direct Vector Access is not supported.

Databricks Apps: up to three apps per account; apps can auto-stop after 24 hours and be restarted.

Lakebase: one project per account, with scale-to-zero behaviour.

Workspace administration: one workspace and one metastore; no account APIs, SSO or SCIM.

Languages and features: no R or Scala, no custom workspace storage, online tables or clean rooms.

Service level: non-commercial use, with no support SLA or guaranteed reliability.

How to conserve quota

Work locally when practical, then deploy to Free Edition regularly.

Deploy once early to confirm the real deployment path.

Use a representative data sample that proves the idea without wasted compute.

Stop what you are not using — idle notebooks, jobs, pipelines, apps and endpoints.

If something stops working

Compute is unavailable	Stop unused resources, reduce the workload, and check whether the daily quota needs time to reset.
A model is unavailable	Select another available pay-per-token model and re-test the response format.
A teammate cannot access the final state	Confirm they were invited to the selected demo workspace and that shared assets live there.
The live path still fails	Show the prepared fallback, explain what the audience is seeing, and continue with the product story.

Submission & demo

Round 1 — idea submission

Closes Tuesday, 6 October 2026, 11:59pm SGT. Round 1 is an idea submission; you do not need a working build yet. Submit through Devpost.

A 1-page concept note or 3-slide pitch deck (PDF) covering: the problem statement you chose, your proposed solution, the datasets you will use, and your intended Databricks architecture.

Team name and full member list: name, institution, course, year and email.

Use the 3-slide template

Every team submits the same three slides — Problem & why it matters, Solution & data, Databricks architecture & impact. An identical structure means judges compare slide 2 with slide 2, instead of hunting through free-form decks. Fill it in, keep it to three slides and export as PDF.

Slide 1 — Problem & why it matters

Slide 2 — Solution & data

Slide 3 — Databricks architecture & impact

Download the template (PowerPoint) View the template (PDF)

Round 2 — Demo Day

For the top 10 teams, due before Demo Day on 27 October 2026.

Tips for a strong demo

Scope aggressively — a polished end-to-end solution on one dataset beats a half-built solution across many.

Tell a story — your dashboard should answer a clear question, not just display data.

Show the pipeline — mentors want to see how data moves from source to insight.

Handle messy data — real datasets have nulls, inconsistencies and edge cases.

Make it actionable — who would use this, and what decision does it inform?

Demo readiness checklist

- [ ] The final deployment owner can sign in and open every required resource.
- [ ] The app, endpoint, SQL warehouse, pipeline and data are ready in the final workspace.
- [ ] The main user journey has been tested from a fresh browser session.
- [ ] No token, password, private key or personal data is visible.
- [ ] The team can explain the problem, user, architecture, Databricks contribution and presenter handoffs.
- [ ] Third-party datasets, libraries, models and templates are credited.
- [ ] A screenshot or short recording is ready for the most failure-prone live step.
- [ ] The final link, repository and submission materials are accessible to judges.

Judging criteria

Both rounds are scored against published criteria. Judges' decisions are final.

Round 1 — idea submission

Criterion	Weight	What judges look for
Problem fit & social impact	30%	Clearly addresses the chosen problem statement, with a real, describable benefit.
Solution quality & originality	30%	A thoughtful approach, not a generic dashboard.
Data & technical feasibility	25%	Named open datasets, a credible Databricks architecture, and buildable in two weeks.
Clarity of submission	15%	Concise, well-structured concept note or deck.

Demo Day — final solution

Criterion	Weight	What judges look for
Social impact	30%	Who benefits, how much, and whether the solution could realistically be adopted.
Technical execution on Databricks	30%	A working end-to-end pipeline: ingestion, transformation, governance, analytics or AI.
Solution & user experience	20%	A usable dashboard, app or interface a non-technical stakeholder could act on.
Data rigour	10%	Sound use of Singapore open data, handling of messy data, and honest limitations.
Presentation	10%	A clear story, a confident live demo and a well-handled Q&A.

Official resources

Databricks Free Edition
Overview and account creation

Foundation Model APIs
Hosted model endpoints and supported clients

Databricks AI Dev Kit
Databricks-specific tools and implementation patterns

data.gov.sg
Every dataset in these problem statements

data.gov.sg Developer Guide
Full API documentation for live data

Self-paced course: Get Started with Databricks: End to End
About 2 hours, no prior experience required (Academy sign-in may be required)

Ready to register?

Registration opens 16 September 2026 at Data+AI World Tour Singapore.
Round 1 submissions close 6 October, 11:59pm SGT.

Register your team



For all queries, contact us at **daisichallenge@databricks.com**

Databricks University Alliance